

PRD-012 — Diagnostic Assessment & Product Introduction

Product: PrepHub V2

Status: Approved

Depends on:

- PRD-001 — Authentication & Onboarding
 - PRD-003 — Billing & Subscriptions
 - PRD-004 — Student Home Dashboard
 - PRD-005 — Practice
 - PRD-006 — Practice Session
 - PRD-007 — Session Review & Results
 - PRD-008 — Progress
 - PRD-010 — Profile & Settings
-

1. Purpose

The Diagnostic Assessment establishes the student's initial PrepHub Score Prediction and provides the starting data needed to personalize future adaptive practice.

Before beginning the diagnostic, students complete a short, visually engaging product introduction that explains how PrepHub works and how students can get the greatest benefit from the platform.

The diagnostic is free for every student, including independent users who have not purchased a subscription.

2. Product Philosophy

The diagnostic should not feel like a traditional placement test.

It should feel like the beginning of a personalized learning journey.

The student should understand that:

- Every answer helps PrepHub understand their current skills.
 - Every completed set helps PrepHub improve future recommendations.
 - Honest effort produces better personalization.
 - Mistakes are useful because they reveal what the student should learn next.
 - Consistent effort matters more than starting ability.
-

3. Goals

The Diagnostic Assessment should:

- Introduce students to PrepHub's learning model.
 - Establish an initial predicted SAT score range.
 - Assess performance across all seven PrepHub categories.
 - Introduce the question-answering and explanation experience.
 - Generate enough initial data to create the next adaptive practice set.
 - Allow students to complete the assessment at their own pace.
 - Provide immediate teaching after every submitted answer.
 - Give all students a meaningful result before requiring payment.
-

4. Non-Goals

The Diagnostic Assessment does not:

- Permanently determine a student's level.
 - Lock future practice into an equal category distribution.
 - Use response time to determine future question selection.
 - Automatically submit when suggested time expires.
 - Prevent students from leaving and returning.
 - Allow students to change an answer after submission.
 - Provide a permanent question-history archive.
 - Require independent users to purchase before seeing their results.
-

5. Post-Signup Flow

Immediately after creating an account, the student enters the PrepHub product introduction.

The intended flow is:

1. Account created
2. Product introduction begins
3. Student chooses:
 - Begin Diagnostic
 - Explore PrepHub
4. Student may explore eligible parts of the platform
5. Practice remains locked until the diagnostic is completed
6. Selecting Practice redirects the student to begin or resume the diagnostic
7. Completing the diagnostic generates the first predicted SAT score range
8. Student reviews results and questions
9. Student continues according to access type

The diagnostic should remain clearly recommended even though students may explore first.

6. Product Introduction

The product introduction is a short, multi-screen onboarding experience.

It should be:

- Visually engaging
- Fast to complete
- Easy to understand
- Mobile-friendly
- Animated where appropriate
- More immersive than a static instructional page
- Skippable only by choosing to explore the platform

The introduction should create excitement and confidence without overwhelming the student with text.

7. Introduction Screen Structure

The introduction should communicate the following ideas across several screens.

Screen 1 — PrepHub Learns With You

Core message:

PrepHub continuously updates your learning experience. Every set you complete helps determine which questions you receive next.

The visual design may show the platform adapting or changing based on student responses.

Screen 2 — Every Answer Matters

Core message:

Try your best on every question. Your answers help PrepHub understand what you know and what you should practice next.

Students should understand that randomly guessing or intentionally answering incorrectly will reduce the quality of their recommendations.

Screen 3 — Complete One Set at a Time

Core message:

For the best results, complete one full set whenever you practice. Each completed set gives PrepHub new information about your progress.

The student may still pause and resume when necessary, but the platform should encourage completing a full set in one study period.

Screen 4 — Your Prediction Keeps Updating

Core message:

After every completed set, PrepHub updates your predicted SAT score range so you can see how your preparation is progressing.

The visual may demonstrate one score range changing into another.

Screen 5 — Learn From Every Question

Core message:

After submitting each answer, you will immediately receive the correct answer, a written explanation, and a video explanation.

The student should understand that explanations are a central part of the PrepHub experience, not an optional afterthought.

Screen 6 — Built for Efficient Progress

Core message:

PrepHub is designed to help you find your weaknesses, learn from mistakes, and improve as efficiently as possible.

The platform should emphasize progression rather than merely completing questions.

Screen 7 — Effort Message

The final introduction screen includes the PrepHub message:

Talent may affect where you begin. Consistent effort determines how far you go.

The screen should end with two actions:

Primary:

Begin Diagnostic

Secondary:

8. Exploring Before the Diagnostic

Students may explore the platform before completing the diagnostic.

They may access:

- Student Dashboard
- Progress
- School Community, when applicable
- Profile & Settings
- Subscription information, when applicable

Practice cannot begin until the diagnostic is completed.

Selecting a Practice action should open or resume the diagnostic instead.

9. Pre-Diagnostic Empty States

Pages that depend on diagnostic data must display intentional empty states.

Dashboard

The Dashboard should explain:

Complete your diagnostic to generate your first PrepHub Score Prediction and personalized practice plan.

Primary action:

Begin Diagnostic

If the diagnostic has already been started:

Resume Diagnostic

Progress

The Progress page should explain:

Your progress journey begins after you complete your diagnostic.

Primary action:

Begin Diagnostic

If already started:

Resume Diagnostic

Practice

The Practice page remains locked.

It should explain:

Complete your diagnostic so PrepHub can create your personalized practice sets.

Primary action:

Begin Diagnostic

If already started:

Resume Diagnostic

10. Diagnostic Structure

The initial diagnostic contains exactly 21 questions.

It assesses seven categories:

1. Reading Comprehension
2. Grammar
3. Vocabulary
4. Algebra

5. Geometry & Trigonometry
6. Advanced Math
7. Problem Solving & Data Analysis

Each category receives exactly three questions during the initial diagnostic.

11. Initial Difficulty Distribution

Each category contains:

- One easy question
- One medium question
- One hard question

This produces the following total distribution:

- Seven easy questions
- Seven medium questions
- Seven hard questions

This balanced structure applies only to the initial diagnostic.

Later adaptive sets may contain:

- More questions from weak categories
- Fewer questions from strong categories
- More easy questions
- More medium questions
- More hard questions

Future distributions depend on the student's completed-set performance.

12. Question Order

Questions may be ordered in a way that creates a smooth and approachable experience.

The initial diagnostic does not need to visibly group questions by category.

The system should avoid making the test feel like seven separate mini-assessments unless doing so improves the experience.

Question order must not reveal internal difficulty labels to students.

Students should not be told whether a question is easy, medium, or hard.

13. Suggested Time Per Question

Every diagnostic question includes its own suggested-time value.

The suggested time is stored as part of the question's content metadata when the question is uploaded into PrepHub.

Examples may include:

- 45 seconds
- 60 seconds
- 90 seconds
- 120 seconds

The interface displays the suggested time for the active question.

14. Question Timer Behavior

The timer measures how long the student spends on the current question.

When the student reaches the question's suggested time:

- The system informs the student that the suggested time has elapsed.
- The student is not forced to submit.
- The student may continue working for as long as needed.
- The question remains fully answerable.
- A correct answer still receives full credit.
- The student is not penalized for exceeding the suggested time.

The notification should be noticeable but not alarming.

Example:

Suggested time reached. You can continue working on this question.

15. Timing and Adaptivity

Question response time does not affect:

- Future category selection
- Future difficulty selection
- Mastery calculations
- Predicted SAT score
- Whether an answer is counted as correct
- The adaptive recommendation algorithm

PrepHub may retain timing information for internal product analysis, but it is not used to evaluate the student's academic ability in V1.

The adaptive system is based on answer performance rather than speed.

16. Pause and Resume

Students may leave the diagnostic and return later.

The system automatically saves:

- Current question position
- Submitted answers
- Unsubmitted answer selections
- Skipped questions
- Remaining unanswered questions
- Diagnostic completion status

When the student returns, PrepHub resumes from the last appropriate position.

If the student selects Practice anywhere in the platform while the diagnostic is incomplete, PrepHub opens the saved diagnostic session.

17. Navigation

Before submitting an answer, students may:

- Move to previous questions
- Move to later questions
- Change selected answers
- Skip questions temporarily
- Return to skipped questions

The interface should clearly identify:

- Current question
 - Answered questions
 - Unanswered questions
 - Skipped questions
 - Submitted questions
-

18. Answer Submission

A question is evaluated only when the student deliberately submits the answer.

After submission:

- The answer is final.
- The student cannot change it.
- The result contributes to diagnostic performance.
- The explanation experience begins.

This rule protects the integrity of the diagnostic and future adaptive recommendations.

Students must not be able to view the correct answer and then revise their submitted response.

19. Immediate Question Feedback

After each answer is submitted, the student receives immediate feedback.

The feedback includes:

- Whether the submitted answer was correct or incorrect
- The correct answer
- A written explanation
- A video explanation, when available

The student should review the explanation before continuing, but watching the entire video is not required to advance.

The feedback experience should emphasize learning rather than punishment.

20. Skipped and Unanswered Questions

Students may skip questions temporarily.

A student cannot complete the diagnostic while required questions remain unanswered.

Before final completion, PrepHub should show a clear review state identifying unanswered questions.

The student must return to and submit an answer for every question.

The system must not silently count skipped questions as incorrect while the diagnostic is still active.

21. Calculator

PrepHub should provide a built-in Desmos calculator for questions where calculator use is permitted.

The availability of the calculator is controlled at the individual-question level.

Each question's metadata should indicate whether the calculator is:

- Allowed
- Not allowed
- Required or recommended, if such a distinction is later needed

The calculator should appear directly within the question interface and should not require students to leave PrepHub.

If a native embedded Desmos experience is not technically available during initial development, PrepHub may use an equivalent approved calculator solution while preserving the intended workflow.

22. Diagnostic Completion

The diagnostic is completed when:

- All 21 questions have been submitted
- No required questions remain unanswered
- The student confirms completion when prompted

After completion, PrepHub:

1. Evaluates the full diagnostic
2. Calculates the student's first predicted SAT score range
3. Records category-level performance
4. Creates the student's next adaptive practice set
5. Opens the Session Review and Results experience

The next adaptive set is generated in the background or immediately after diagnostic completion so it is available when the student receives practice access.

23. Diagnostic Results

After completion, the student receives the standard completed-set results experience.

The results include:

- Initial PrepHub Score Prediction range
- Session breakdown
- Category breakdown
- Questions correct
- Total questions
- Accuracy
- Average time per question
- Total session time
- Review of all diagnostic questions

The prediction is displayed as a range rather than a single exact score.

24. Diagnostic Question Review

Students may review diagnostic questions:

- Immediately after submitting each question
- From the completed diagnostic Session Review page

The Session Review allows access to:

- Original question
- Student's submitted answer
- Correct answer
- Written explanation
- Video explanation

Once the student leaves the completed-set review experience, the individual diagnostic questions are not permanently available through a historical question archive.

The resulting prediction and long-term progress data remain available.

25. No Traditional Diagnostic Retake

PrepHub does not provide a normal "Retake Diagnostic" action.

The initial diagnostic establishes the first baseline.

Afterward, every completed adaptive practice set continues evaluating the student and updating:

- Category understanding
- Future question selection
- Difficulty distribution
- Predicted SAT score range

In this sense, every subsequent set acts as a continued assessment.

Exceptional resets caused by technical problems or invalid attempts may be handled by PrepHub support or authorized internal staff.

26. Access After Completion

Every student can complete the diagnostic and view the complete result before payment is required.

School-Sponsored Students

After reviewing results, school-sponsored students may:

- Continue to the Dashboard
- Begin the next adaptive practice set
- Use all features included in school access

Primary continuation action:

Go to Dashboard

A secondary action may allow:

Continue Practice

Paid Individual Students

Paid individual students follow the same continuation flow as school-sponsored students.

They may continue to the Dashboard or begin the next adaptive set.

Unpaid Individual Students

Unpaid individual students receive:

- Their full predicted SAT score range
- Diagnostic session statistics
- Category breakdown

- Immediate question explanations
- Completed-set question review

After the result and review experience, PrepHub directs the student to subscription selection before allowing another practice set.

The student may continue viewing eligible non-practice areas, but further practice remains locked.

Primary action:

Choose a Plan

The student's diagnostic result and progress must remain saved if they leave without purchasing.

27. Diagnostic Access States

The system should recognize the following diagnostic states:

- Not Started
- In Progress
- Completed
- Completed — Practice Locked Pending Subscription
- Exceptional Reset Required

These states control:

- Dashboard empty-state messaging
 - Progress empty-state messaging
 - Practice access
 - Resume behavior
 - Subscription prompts
 - Result availability
-

28. Visual Design

The diagnostic should feel modern, focused, and less intimidating than a traditional standardized test.

The interface should prioritize:

- Readable question content
- Clear answer choices
- Easy navigation
- Visible but non-stressful timing
- Smooth transitions into explanations
- Automatic saving
- Clear progress through the 21 questions

The product introduction should feel especially polished because it forms the student's first impression of the PrepHub learning experience.

Animations should support comprehension and excitement without slowing the user down.

29. Acceptance Criteria

Product Introduction

- Appears immediately after signup.
- Uses a multi-screen format.
- Explains continuous adaptation.
- Explains why honest effort matters.
- Encourages completing one set at a time.
- Explains predicted score updates.
- Explains immediate written and video feedback.
- Includes the approved effort message.
- Provides Begin Diagnostic and Explore PrepHub actions.

Exploration

- Students may explore before completing the diagnostic.
- Dashboard displays a diagnostic empty state.
- Progress displays a diagnostic empty state.
- Practice remains locked.
- Practice actions begin or resume the diagnostic.

Structure

- Diagnostic contains exactly 21 questions.
- Covers all seven categories.
- Includes three questions per category.
- Includes one easy, one medium, and one hard question per category.

Timing

- Every question has a suggested-time value.
- The interface displays question-level timing.
- Students are notified when suggested time is reached.
- Time expiration does not force submission.
- Students receive full credit for correct answers submitted after suggested time.
- Response time does not affect adaptivity or predicted score.

Session Persistence

- Students may pause and resume.
- Progress saves automatically.
- Submitted answers remain locked.
- Unsubmitted selections and skipped questions are preserved.

Navigation

- Students may move backward before submission.
- Students may change unsubmitted answers.
- Students may skip questions temporarily.
- Students cannot finish with unanswered questions.

Feedback

- Submitted answers cannot be changed.
- Students immediately see correct or incorrect status.
- Students receive the correct answer.
- Students receive a written explanation.
- Students receive a video explanation when available.

Calculator

- Calculator availability is controlled by question metadata.
- Permitted questions provide an integrated calculator experience.
- Desmos is the preferred calculator solution.

Completion

- Completion generates the first predicted SAT score range.
- Completion records category performance.
- Completion generates the next adaptive set.
- Completion opens the standard Session Review experience.

Review

- Students can review questions immediately after answering.
- Students can review all questions during completed-set review.
- Questions are not permanently available after leaving the review experience.

Access

- The diagnostic is free for every user.
- Unpaid users see full diagnostic results before purchase.
- Unpaid users cannot begin another set without subscribing.
- School-sponsored and paid users may continue practicing.
- Diagnostic results remain saved if an unpaid student does not purchase.

Overall Experience

After completing the diagnostic, students should understand:

- How PrepHub works
- Why every answer matters
- Their initial predicted SAT score range
- Which categories currently need improvement
- How explanations help them learn
- That future sets will continue adapting to their performance
- What action they should take next